

Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

Example: The expression 1.5^{3t+2} is equivalent to the expression $2.25(1.5)^{3t}$ which is equivalent to $2.25(3.375)^t$.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.1.1, MA.912.AR.1.3,
- MA.912.AR.1.4, MA.912.AR.1.7
- MA.912.AR.5.3
- MA.912.FL.3.2

Terms from the K-12 Glossary

- Base
- Expression
- Exponent

Vertical Alignment

Previous Benchmarks

- MA.8.AR.1.1

Next Benchmarks

- MA.912.NSO.1.7

Purpose and Instructional Strategies

In grade 8, students generated equivalent algebraic expressions using the Laws of Exponents with integer exponents. In Algebra I, students expand this work to include rational-number exponents. In later courses, students extend the Laws of Exponents to algebraic expressions with logarithms.

- Instruction includes using the terms Laws of Exponents and properties of exponents interchangeably.
- Instruction includes student discovery of the patterns and the connection to mathematical operations (*MTR.5.1*).
- Students should be able to fluently apply the Laws of Exponents in both directions.
 - For example, students should recognize that a^6 is the quantity $(a^3)^2$; this may be helpful when students are factoring a difference of squares.
- When generating equivalent expressions, students should be encouraged to approach from different entry points and discuss how they are different but equivalent strategies (*MTR.2.1*).
- The expectation for this benchmark does not include the conversion of an algebraic expression from exponential form to radical form and from radical form to exponential form.

Common Misconceptions or Errors

- Students may not understand the difference between an expression and an equation.
- Students may not have fully mastered the Laws of Exponents and understand the mathematical connections between the bases and the exponents.
- Student may believe that with the introduction of variables, the properties of exponents differ from numerical expressions.

Strategies to Support Tiered Instruction

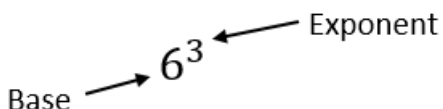
- Instruction includes the opportunity to distinguish between an expression and an equation. These should be captured in a math journal.
 - For example, when generating equivalent expressions, place an equal sign in between the expressions and label each expression and the equation.

$$\begin{array}{ccc}
 & \text{equation} & \\
 \swarrow & & \searrow \\
 1.5^{3t+2} = 2.25(1.5)^{3t} & & \\
 \swarrow & & \searrow \\
 \text{expression} & & \text{expression}
 \end{array}$$

- Instruction provides opportunities to write each term in expanded form first and then use Laws of Exponents to combine like factors. It may also be helpful to chunk each step.
 - For example, to rewrite the expression $(8x^3)^2$ with one exponent, write out $(8)(x)(x)(x)(8)(x)(x)(x)$ and then use the commutative property to write $(8)(8)(x)(x)(x)(x)(x)(x) = 64x^6$.
- Teacher provides instruction for problems that require multiple applications of the Laws of Exponents by chunking the steps so that students are applying one property at each time and explaining the property applied. Each time ask students to identify the property of exponents that they applied.
- Teacher provides students side-by-side problems, one with variable bases and the other choosing a value for the variable. As students work through the problems, ask them about the similarities in the problem-solving process.
 - For example, teacher can model generating equivalent expressions like the ones below.

$\left(\frac{5(3)^4}{4(3)^2}\right)^{0.5} =$	$\left(\frac{5(x)^4}{4(x)^2}\right)^{0.5} =$
$\frac{5^{0.5}3^{(4 \cdot 0.5)}}{4^{0.5}3^{(2 \cdot 0.5)}} =$	$\frac{5^{0.5}x^{(4 \cdot 0.5)}}{4^{0.5}x^{(2 \cdot 0.5)}} =$
$\frac{5^{0.5}(3^2)}{2(3^1)} =$	$\frac{5^{0.5}(x^2)}{2(x^1)} =$
$\frac{5^{0.5}(3^{2-1})}{2} =$	$\frac{5^{0.5}(x^{2-1})}{2} =$
$\frac{5^{0.5}(3^1)}{2} =$	$\frac{5^{0.5}(x^1)}{2} =$
$\frac{3\sqrt{5}}{2}$	$\frac{x\sqrt{5}}{2}$

- Teacher provides a review of the relationship between the base and the exponent by modeling an example of operations using a base and exponent.
 - For example, determine the numerical value of 6^3 .



 6^3 which is equivalent to $6 \cdot 6 \cdot 6$ which is equivalent to 216.

Instructional Tasks

Instructional Task 1 (MTR.3.1, MTR.5.1)

Given the function $h(x) = 10^{0.2x}$, what is the rate of growth or decay?

Instructional Task 2 (MTR.3.1, MTR.4.1)

Part A. Write the algebraic expression $\left(\frac{6x^{\frac{2}{7}}y^{-4}z^0}{9x^2y^5z^{-8}}\right)^3$ as an equivalent expression where each variable only appears once.

Part B. Compare your method of simplifying with a partner.

Instructional Items

Instructional Item 1

Given the algebraic expression 2.3^{2t-1} , create an equivalent expression.

Instructional Item 2

Use the properties of exponents to create an equivalent expression for the given expression shown below with no variables in the denominator.

$$(64x^2)^{-\frac{1}{6}}(32x^5)^{-\frac{2}{5}}$$

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*